

# ThermoShelter AI — Final Prototype Blueprint & Thermal Analysis

SIH26051 | Team Agnit | Software-based area-specific shelter thermal-comfort prototype

## 1. Shelter Specification

| Parameter    | Value                |
|--------------|----------------------|
| Geometry     | Chamfered Corner     |
| Dimensions   | 4.00 × 3.00 × 2.50 m |
| Corner angle | 90°                  |
| Material     | XPS                  |
| Thickness    | 0.150 m              |
| Location     | 28.613900, 77.209000 |

## 2. Climate & Thermal Results

| Metric                       | Result               |
|------------------------------|----------------------|
| Environment source           | Manual climate input |
| Outdoor temperature          | -10.00 °C            |
| RH                           | 50.0%                |
| R-value                      | 4.412 m²K/W          |
| U-value                      | 0.227 W/m²K          |
| Net thermal load             | -351.4 W             |
| Predicted indoor temperature | 22.50 °C             |
| PMV                          | +0.15                |
| PPD                          | 5.5%                 |
| Thermal status               | OPTIMAL COMFORT      |
| Heatmap proxy range          | 22.4–22.7 °C         |

## 3. PMV Model Inputs

Air temperature=22.50 °C; mean radiant temperature=21.98 °C; RH=50.0%; air speed=0.10 m/s; activity=1.2 met; clothing=1.0 clo.

## 4. Recommendations

**THERMAL RESPONSE** — Current PMV is within the selected ±0.5 screening band; verify across seasonal/extreme conditions.

**COLD-CLIMATE ADAPTATION** — Envelope resistance and uncontrolled ventilation deserve priority; review geometry for thermal-bridge-sensitive areas.

**MATERIAL SCREENING** — Current screening selects XPS at 0.10 m by the prototype ranking. R=2.941 m²K/W, predicted indoor=22.5 °C. Verify properties and structural/moisture/fire requirements before engineering use.

## 5. Research & Standards Basis

Development basis includes the five supplied shelter/thermal-comfort studies, Fanger PMV/PPD methodology, ASHRAE Standard 55 and ISO 7730. The prototype does not claim standards certification, experimental validation or validated CFD. Material property values must be verified against authoritative project-approved data before engineering deployment.

## 6. Limitations

The heatmap is a simplified spatial visualization proxy. The thermal solver is a simplified steady-state analytical model with an explicit internal-gain assumption and is not a complete transient shelter simulation. Corner-angle visualization does not constitute validated thermal-bridge simulation. Device location depends on operating-system support and permission.